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Statistical Testing on E-Commerce Dataset

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Introduction & Dataset Overview

This report details the practical implementation of five statistical tests on the Online Retail "E-Commerce Data" dataset. The dataset consists of invoice-level transactions detailing product quantities, unit prices, customer IDs, and geographical locations. During initial data preparation, the dataset was strictly filtered to include only valid transactions (i.e., rows where both quantity > 0 and unit price > 0), and a new target variable, `revenue`, was engineered as the mathematical product of quantity and unit price. Exploratory data analysis visually revealed that the numeric data is heavily right-skewed. The retail environment is characterized by a massive volume of low-cost, small-quantity purchases forming a dense lower bound, accompanied by a long tail of extreme, high-value wholesale transactions.

Test 1: Normality of Revenue

Question: Does the distribution of line-item revenue resemble a normal distribution?
Variables Used: `revenue` (Random sample size: 3000)
To evaluate the assumption of normality, a Shapiro-Wilk test was conducted on a randomized sample of 3,000 observations drawn from the `revenue` variable. The test yielded a test statistic of 0.3038 and a p-value of 5.9658×10^{-75}. Because the p-value is significantly less than the standard threshold ($\alpha = 0.05$), we reject the null hypothesis. The revenue data strictly violates the normality assumption, demonstrating profound right-skewness. In the context of e-commerce, this mathematically proves that customer purchasing behavior is not uniformly distributed around an average; instead, the vast majority of items generate minimal revenue, heavily skewed by rare but massive bulk-order outliers.

Test 2: Association Between Quantity and Unit Price

Question: Is there an association between the quantity ordered and the unit price?
Variables Used: `quantity`, `unit_price`

A Spearman rank correlation test was utilized to assess the monotonic relationship between quantity and unit price. The test produced a correlation coefficient (r_s) of -0.4032 and a p-value of < 0.0001. With a p-value strictly below 0.05, the null hypothesis is rejected. The negative sign of the correlation coefficient confirms a statistically significant, inverse monotonic relationship between the two metrics. In practical business terms, this reflects a standard consumer trend: as the unit price of an item increases, the quantity ordered by a customer tends to decrease. Conversely, low-cost or cheaper items are far more likely to be purchased in high bulk quantities.

Test 3: Revenue Differences - UK vs. Rest of World

Question: Do line-item revenues differ on average between the United Kingdom and all other countries combined?
Variables Used: `revenue`, `Country`
An independent two-sample t-test (specifically Welch's t-test, not assuming equal variances) was conducted to compare the mean revenue of transactions originating in the UK against those from the rest of the world. UK transactions exhibited a mean revenue of £16.86, while the combined international group averaged £35.62. The test calculated a t-statistic of -30.8258 and a p-value of 4.9985×10^{-208}. By rejecting the null hypothesis ($p < 0.05$), we conclude there is a highly significant difference in average transaction values. While the domestic UK market dominates total transaction volume, international line items yield significantly higher average revenues, likely driven by international shipping structures that disincentivize small single-item orders in favor of larger wholesale purchases.

Test 4: Revenue Across Top Transaction Countries

Question: Among the top countries by number of transactions, are the mean revenues the same?
Variables Used: `revenue`, `Country` (Filtered to Top 5 by volume)
A one-way Analysis of Variance (ANOVA) was employed to compare the mean line-item revenue across the top five highest-volume countries in the dataset. The statistical test produced an F-statistic of 10.2228 and a p-value of 2.8330×10^{-8}. Because the p-value is below the 0.05 significance level, we strongly reject the null hypothesis, concluding that mean revenue is not uniform across these primary regions. This underscores significant geographical variance in consumer behavior; certain top-tier countries consistently drive higher average transaction values than others, indicating distinct regional market compositions (e.g., B2B versus B2C concentrations).

Test 5: Country and High-Value Purchases

Question: Is the proportion of high-value purchases the same across the main countries?
Variables Used: `revenue` (Threshold set at the 99th percentile), `Country`
To evaluate categorical independence, a Chi-Square test was performed. Individual transactions were classified as "high-value" if their revenue met or exceeded the 99th percentile of the dataset. The cross-tabulation across the top five countries yielded a Chi-Square

statistic of 358.1028, with 4 degrees of freedom, and a p-value of 3.1214×10^{-76}. Rejecting the null hypothesis ($p < 0.05$) mathematically proves that the proportion of high-value line items is highly dependent on the transaction's country of origin. This indicates that extreme "whale" clients or wholesale distributors are disproportionately concentrated in specific international markets rather than being evenly distributed.

Closing Reflection

The analytical approach undertaken in this project provided robust, data-driven insights into the e-commerce dataset, successfully capturing the skewed nature of online retail transactions. The non-parametric tests—specifically the Spearman rank correlation and the Chi-Square test of independence—were remarkably appropriate for this dataset. They perform exceptionally well despite the extreme presence of outliers and the non-normal distribution of the financial metrics.

Conversely, the fundamental assumptions underlying the parametric tests implemented in Phase I (the two-sample t-test and the one-way ANOVA) are highly questionable in this context. Both methodologies mathematically assume that the underlying data within each group follows a normal distribution. As definitively proven by the Shapiro-Wilk test (Test 1), the `revenue` variable severely violates this normality assumption. While the enormous sample size affords some leniency via the Central Limit Theorem to validate the comparisons of sample means, the extreme variance and pronounced right-skewness strongly suggest that non-parametric alternatives (such as the Mann-Whitney U test for two groups, or the Kruskal-Wallis test for multiple groups) would offer a far more statistically rigorous methodology for this specific retail environment.